

Normalized Cuts and Cheeger's Inequality

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Image Segmentation

- Given an image, identify relevant regions, large and internally coherent
- Clustering is an unsupervised problem, and very useful.



Figure: Image segmentation: Image segmentation identifies important regions of the image. (a) the input image (grayscale). (b-d) shows the mask of three segments detected by an image segmenter. We see the head, the body of the subject , and the background.

Primer on Laplacians I

- For a graph G , Edge-Incidence matrix $K \in \mathbb{R}^{|E| \times |V|}$

$$K_{eu} = \begin{cases} \sqrt{w_e} & e = (u, v), u < v \\ \sqrt{-w_e} & e = (u, v), u > v \\ 0 & \text{otherwise} \end{cases}$$

- K is the gradient operator on vertex-valued functions

$$(K\mathbf{f})_{uv} = \sqrt{w_{uv}}(f_u - f_v)$$

- K^T is the diverse operator on edge-valued vector fields

$$(K^T \mathbf{g})_u = \sum_{e \in E} \sqrt{w_e} g_e$$

Primer on Laplacians II

- For the *characteristic vector* of a set S , $(\mathcal{X}_S)_u = 1_{u \in S}$, then

$$\|K\mathcal{X}_S\|_2^2 = \sum_{e \in E(S)} (0)^2 + \sum_{e \in E(\bar{S})} (0)^2 + \sum_{e \in \partial S} (\pm\sqrt{w_e})^2 = w(\partial S)$$

- Laplacian is $L = K^T K = D - A \in \mathbb{R}^{|V| \times |V|}$, $D = \text{diag}(A\mathbf{1}_n) = \text{diag}(\mathbf{d})$
- Normalized adjacency and Laplacian

$$\tilde{A} = D^{-1/2} A D^{-1/2} \quad \tilde{L} = D^{-1/2} L D^{-1/2} = I - \tilde{A}$$

- $\mathbf{x}^T L \mathbf{x} = \|K\mathbf{x}\|_2^2 = \sum_{ij \in E} w_{ij} (x_i - x_j)^2$

Spectral clustering I



Figure: Normalized versus standard mincut segmentation: (a) a toy image with four circles (background masked out). (b) using spectral clustering on the four circles correctly identifies circles. (c) using mincut clustering on the four circles finds 3 segments of size 1. Trivial mincuts suggest a need to penalization for small segments.

Spectral clustering II

- Normalized cuts penalizes large boundaries and small segments:

$$\text{vol}(S) = \sum_{u \in S} \sum_{v \in V} w_{uv} \quad \sigma(S_1, \dots, S_k) = \sum_{\ell=1}^k \frac{w(\partial S_\ell)}{\text{vol}(S_\ell)}$$

- Rewrite σ in terms of L and D as Rayleigh-quotient:

$$\sigma(S_1, \dots, S_k) = \sum_{\ell=1}^k \frac{\mathcal{X}_{S_\ell}^T L \mathcal{X}_{S_\ell}}{\mathcal{X}_{S_\ell}^T D \mathcal{X}_{S_\ell}}$$

- Assignment weighted matrix $H \in \mathbb{R}^{n \times k}$, row i is assignment of vertex i

$$h_{ij} = 1_{\{i \in S_j\}} \frac{1}{\sqrt{\text{vol}(S_j)}}$$

Spectral clustering III

- Write $\sigma(G)$ in terms of a discrete matrix optimization problem

$$\begin{aligned}\mathbf{h}_j^T L \mathbf{h}_j &= \frac{\mathcal{X}_{S_j}^T L \mathcal{X}_{S_j}}{\text{vol}(S_j)}, & \mathbf{h}_j^T D \mathbf{h}_j &= \frac{\mathcal{X}_{S_j}^T D \mathcal{X}_{S_j}}{\text{vol}(S_j)} = 1 \\ \implies \sigma(S_1, \dots, S_k) &= \text{Tr}(H^T L H), H^T D H = I_k\end{aligned}$$

- Change of variables to normalized Laplacian

$$J = D^{1/2} H \implies \text{Tr}(H^T L H) = \text{Tr}(J^T \tilde{D} J), J^T J = I_k$$

- SVD of $\tilde{L}$ implies J columns are k smallest eigenvectors of $\tilde{L}$ (singular K).
- Rounding typically done via k -means on spectral embedding.

Spectral clustering IV

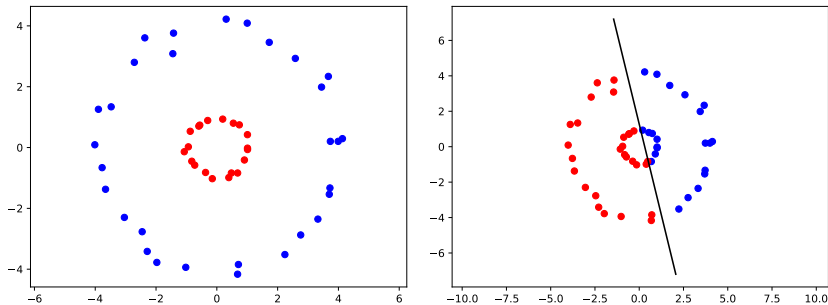


Figure: Spectral clustering: Spectral embedding is able to find clusters based on connectivity. (a), the k-means clustering of Spectral embedding. (b), the k-means clustering using Euclidean embedding.

Cheeger's Inequality (Easy Direction)

- $k = 2$, finding S is a 1-D problem ($S_2 = \bar{S}_1$); $\sigma(G) = \min_S \sigma(S, \bar{S})$.
- Write $\lambda_1 \leq \dots \leq \lambda_n$ eigenvalues, $\mathbf{v}_1, \dots, \mathbf{v}_n$ eigenvectors of $\tilde{L}$
- $\lambda_1 = 0$, as $L\mathbf{1}_n = (D - A)\mathbf{1}_n = \mathbf{d} - \mathbf{d} = \mathbf{0}$
- $\mathbf{v}_1 = \frac{1}{\sqrt{n}}\mathbf{1}_n$ independent of G , and we can discard it
- Relaxed objectives gives inequality in terms of λ_2 :

$$\sigma(G) = \min_{J \text{ assignment}, J^T J = I_k} \text{Tr}(J^T L J) \geq \min_{J^T J = I_k} \text{Tr}(J^T L J) = \lambda_2$$

Cheeger's Inequality (Hard Direction) I

- Rounding a solution via sweep-cuts of $\mathbf{v}_2$:
 - ▶ Compute $\mathbf{x}$ 2nd eigenvector of $\tilde{L}$.
 - ▶ Return best cut among sweeps $\{S_\tau = \{v \in V : \mathbf{x}(v) \leq \tau\}\}$
- WLOG, assume that $\mathbf{x}$ is ascending and $x_1^2 + x_n^2 = 1$.
- For $Y \neq 0$, then considering sign of $Z = X - \text{Ev}(X)Y / \text{Ev}(Y)$,

$$\Pr\left\{\frac{X}{Y} \leq \frac{\text{Ev}(X)}{\text{Ev}(Y)}\right\} > 0$$

- Analysis by defining probability distribution on τ , and showing

$$\text{Ev}(\min(\text{vol}(S_\tau), \text{vol}(\bar{S}_\tau))) = \mathbf{x}^T D \mathbf{x}$$

$$\text{Ev}(|\partial S_\tau|) \leq \sqrt{\mathbf{x}^T \tilde{L} \mathbf{x}} \sqrt{2 \mathbf{x}^T D \mathbf{x}} = \sqrt{2 \lambda_2} \mathbf{x}^T D \mathbf{x}$$

Cheeger's Inequality (Hard Direction) II

- Let density $f(\tau = t) = 2|t|$ (squared mass)
- First bound by noting smaller S determined by zero point of $\mathbf{x}$ ($\mathbf{x} \perp \mathbf{d}$)

$$\Pr\{u \in \arg \min(\text{vol}(S), \text{vol}(\bar{S}))\} = \begin{cases} \tau \in [x(u), 0] & x(u) < 0 \\ \tau \in [0, x(u)] & x(u) > 0 \end{cases}$$

$$\implies \text{Ev}(\min(\text{vol}(S), \text{vol}(\bar{S}))) = \sum_u \deg(u)x(u)^2 = \mathbf{x}^T D \mathbf{x}$$

Cheeger's Inequality (Hard Direction) III

- Second bound with Cauchy-Schwartz to connect ℓ_1 and ℓ_2 norms,

$$\begin{aligned}\Pr\{uv \in E(S, \bar{S})\} &= \begin{cases} |x(u)^2 - x(v)^2| & \text{different signs} \\ x(u)^2 + x(v)^2 & \text{same signs} \end{cases} \\ &\leq |x(u) - x(v)|(|x(u)| + |x(v)|) \\ \implies \mathbb{E}v(|\partial S|) &\leq \sum_{uv \in E} |x(u) - x(v)|(|x(u)| + |x(v)|) \\ &\leq \sqrt{\sum_{uv \in E} w(uv)(x(u) - x(v))^2} \sqrt{\sum_{uv \in E} w(uv)(|x(u)| + |x(v)|)^2} \\ &\leq \sqrt{\mathbf{x}^T L \mathbf{x}} \sqrt{2 \mathbf{x}^T D \mathbf{x}}\end{aligned}$$

Cheeger's Inequality

- We showed that $\lambda_2 \leq \sigma(G) \leq 2\sqrt{2\lambda_2}$
- In terms of the conductance of a graph, $\varphi(S) = \frac{w(\partial S)}{\min(\text{vol}(S), \text{vol}(V-S))}$

$$\frac{\lambda_2}{2} \leq \varphi(G) \leq \sqrt{2\lambda_2}$$

- Establishes equivalence between spectral expanders and edge expanders
- Useful in bounding mixing times of random walks, $(I - \tilde{L})$ is the deviation from stationary distribution